

Gradient Boosting Explained | How it Works

1. Core Idea

- Gradient Boosting is an **ensemble method** that builds models **sequentially**.
- Each new model tries to **correct the errors (residuals)** of the previous ones.
- It uses **gradient descent** to minimize a loss function step by step.

👉 Think of it like:

“I build a weak model → check where it fails → train the next model to fix those mistakes → repeat.”

2. Step-by-Step Process

1. *Start with an initial prediction*

1. For regression → could be the mean of target values.
2. For classification → could be log odds.

2. *Compute residuals (errors)*

1. Residual = (actual value – predicted value).
2. These residuals show what’s left to be explained.

3. *Fit a weak learner (usually a decision tree)*

1. Train the tree to predict these residuals.
2. The tree learns “where the model went wrong.”

4. *Update predictions*

- Add the new tree’s predictions (scaled by a learning rate) to the previous model’s prediction.

$$\hat{y}_{new} = \hat{y}_{old} + \eta \cdot h(x)$$

where:

- η = learning rate (small step size).
- $h(x)$ = weak learner (new tree).

5. **Repeat** until reaching the number of trees or convergence.

3. Why It’s Powerful

- Each tree **fixes the mistakes** of the prior ones.
 - Uses **gradient descent optimization** to reduce the loss function.
 - Flexible: works with many loss functions (squared error, log loss, etc.).
 - Great balance between bias and variance.
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➤ Main Difference: AdaBoost vs Gradient Boosting

Aspect	AdaBoost	Gradient Boosting
Error focus	Adjusts weights of samples : misclassified points get higher weight.	Fits new learners to the residuals (errors) using gradient descent.
How next model is trained	Learns by focusing more on “hard” samples.	Learns by minimizing the loss function step-by-step.
Mathematical basis	Reweights data → misclassified samples matter more.	Gradient descent on loss function → optimize directly.
Weak learner	Often very shallow trees (stumps).	Usually slightly deeper trees (not just stumps).
Interpretation	“Pay more attention to hard examples.”	“Correct the residual errors using gradient descent.”

👉 In short:

- **AdaBoost**: reweights data points → focus on hard cases.
- **Gradient Boosting**: fits new models on residuals → gradient descent on loss function.

✅ Summary:

Gradient Boosting is a boosting method where each new model corrects previous errors by fitting to residuals, optimized via gradient descent. The main difference from AdaBoost is **how errors are handled**: AdaBoost adjusts sample weights, while Gradient Boosting directly minimizes the loss function using gradients.
